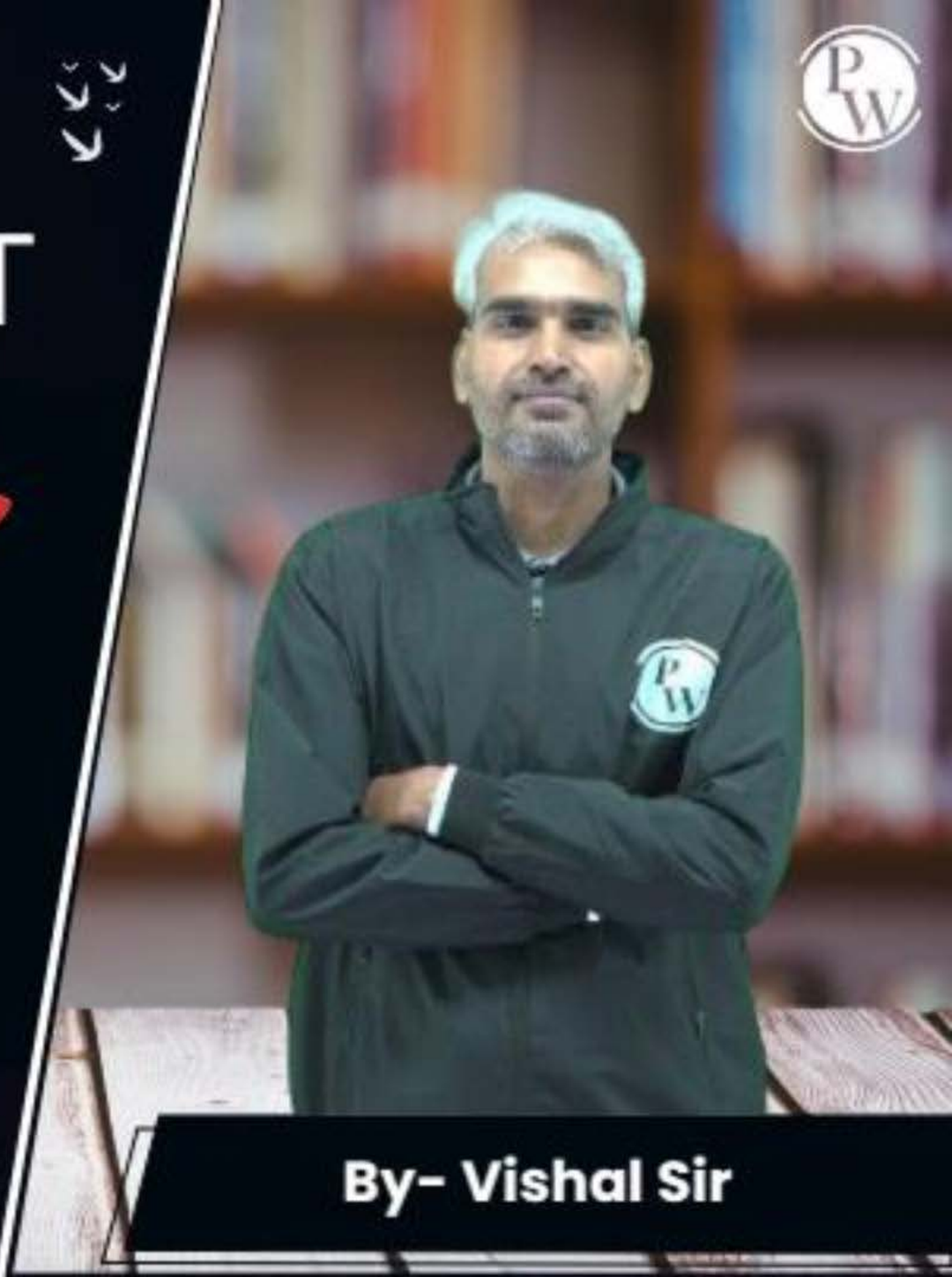
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 12



By- Vishal Sir

Recap of Previous Lecture



Topic — Connectivity

Topic — Spanning tree

Slide

Topics to be Covered



Topic — Euler train, Euler circuit, and traversable graph

Topic — Hamiltonian path and Hamiltonian cycle

Topic — Distance, Eccentricity, Diameter, Radius and Girth

Topic — n-cube graph

Topic — Clique

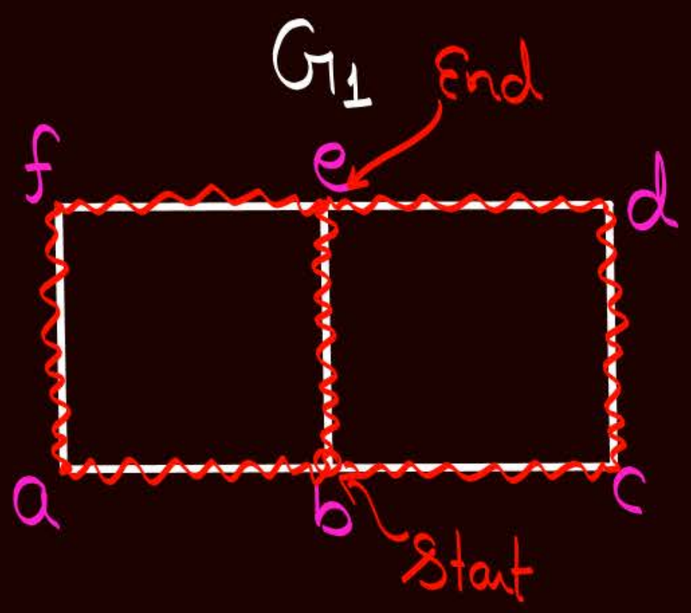
Slide



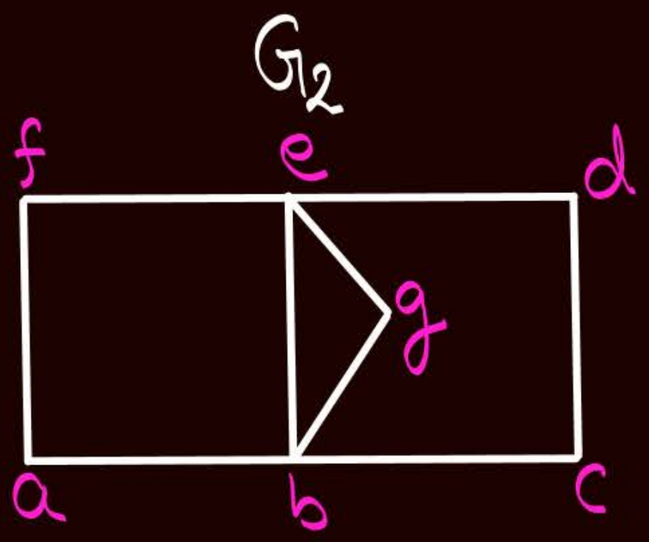
Topic : Euler trail / Euler path

i.e. Start & End vertex are different

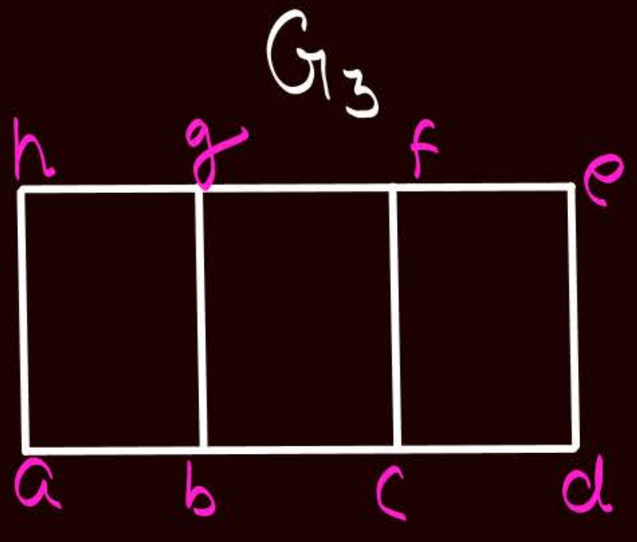
- Let G be a Connected graph, an open trail that visit every edge of the graph exactly once and every vertex of the graph at least once is called an Euler trail / Euler path



Euler trail (open) Exist in graph G_1



Euler trail (open) does not Exist in graph G_2



Euler trail (open) does not Exist in graph G_3

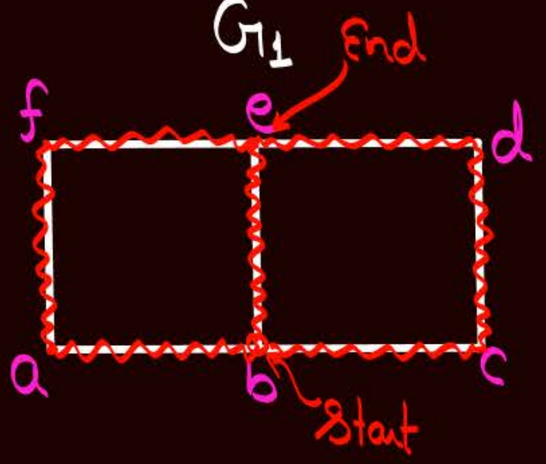


Topic : Euler circuit

Closed

Start & End vertex must be same

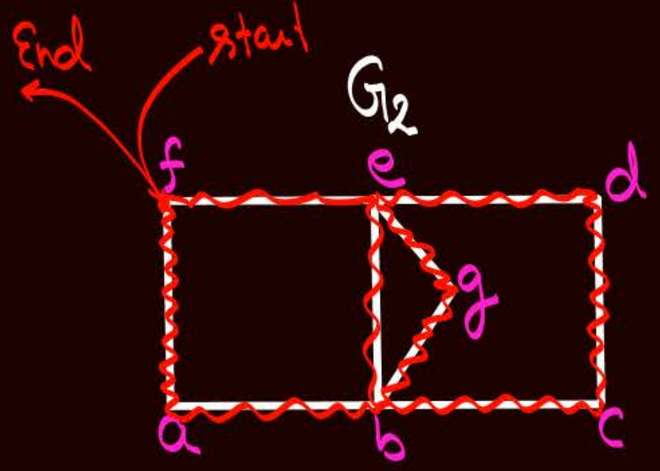
In a connected graph G, a closed trail that visit every edge of the graph exactly once and every vertex of the graph at least once, is called an "Euler Circuit"



Euler trail (open) Exist in graph G_1

But Euler circuit does not Exist in G_1

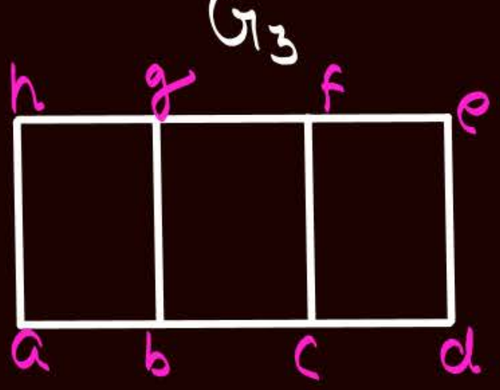
In graph G_1 there Are exactly two Vertices of odd degree



Euler trail (open) does not Exist in graph G_2

But Euler Circuit Exist in G_2

In graph G_2 there Are '0' vertices of odd degree



Euler trail (open) does not Exist in graph G_3

& Euler Circuit also does not exist in G_3

In graph G_3 there Are '4' {i.e. more than '2'} Vertices of odd degree

Note:

① If graph G is Connected graph, and there are no Vertices of odd degree, then Euler Circuit will exist in graph G. but no Euler trail

② If graph G is Connected graph, and there are exactly two Vertices of odd degree, then Euler Circuit will not exist in graph G. but Euler trail exist.

③ If graph G is Connected graph, and there are more than two Vertices of odd degree, then no Euler Circuit and no Euler trail will exist in G.

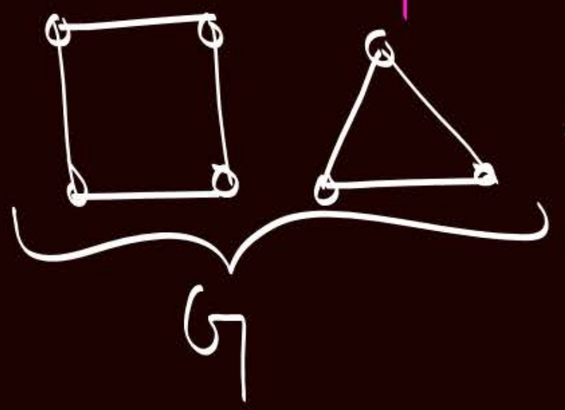
Note: If graph G is a disconnected graph, then there will be no Euler train and no Euler Circuit.

→ Check whether the statement is true or false,

- In a graph G if all vertices are of even degree then Euler circuit Exists in graph G.

→ False, because they did not specify that the graph is Connected.

Eg:



No vertex of odd degree, but no Euler Circuit



Topic : Eulerian graph / Euler graph

A graph G is an Eulerian graph if and only if Euler Circuit exists in graph G.

(or)

A graph G is an Eulerian graph if and only if graph G is a Connected graph and degree of each Vertex is Even.



Topic : Traversable graph

A graph G is called traversable if and only if Euler circuit or Euler trail exists in graph G.

(or)

A graph G is called traversable if and only if, graph G is a Connected graph and number of vertices with odd degree are '0' or '2'.



Topic : Hamiltonian path

(open)

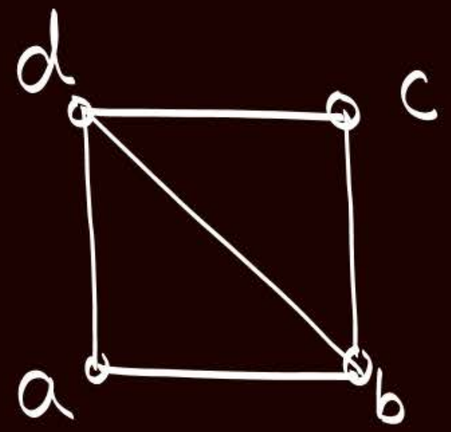
Start & End vertex must be different

An open path that visit Each vertex of the graph Exactly once { we may skip some of the edges } is called Hamiltonian Path

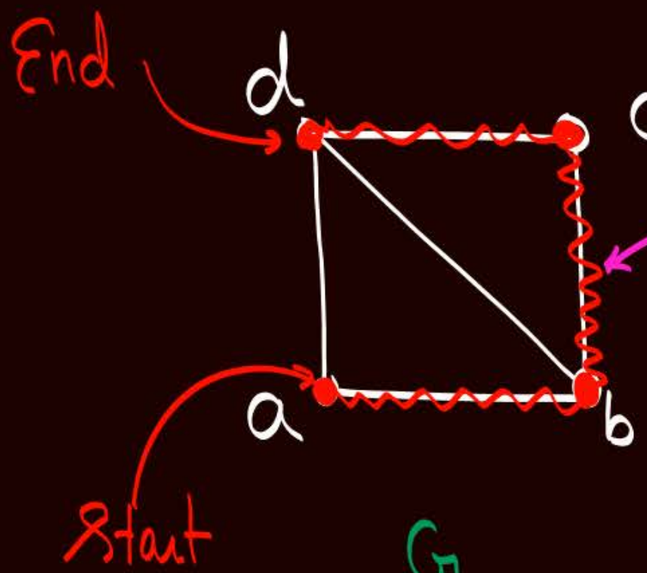
(or)

An open path that visit Each vertex of the graph Exactly once and Each edge of the graph at most once is called Hamiltonian path

eg



G



G

in this path we have visited Every Vertex Exactly once

∴ Hamiltonian Path Exist in graph G.



Topic : Hamiltonian circuit

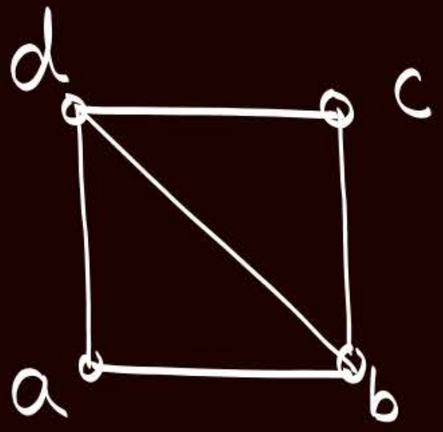
Closed

(Hamiltonian (or) Cycle)

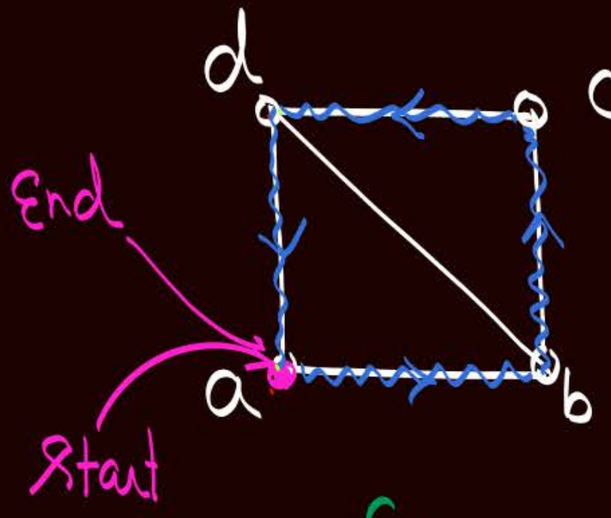
Start & End vertex must be same

A Closed path that visit Each vertex of the graph Exactly once Except for the start vertex {it will be visited twice}, and each edge of the graph at most once is called Hamiltonian circuit.

eg



G



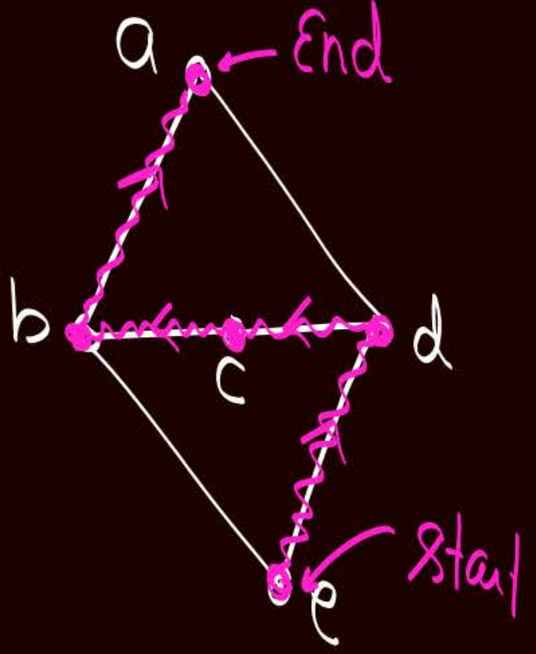
G

Hamiltonian Circuit Exist in graph G

Note :- If Hamiltonian Circuit Exists in the graph G, then Hamiltonian path will also exist in graph G.

But If Hamiltonian Path exist in graph G, then Hamiltonian Circuit need not exist in graph G
(may or may not)

e.g.



G

Hamiltonian path (open) Exist in graph G,
but Hamiltonian Circuit does not exist in graph G.



Topic : Hamiltonian graph

A graph G is called Hamiltonian graph if and only if Hamiltonian circuit (Closed) Exist in graph G



← No Hamiltonian Path & No Hamiltonian Cycle

Note:- There is no trick to check whether an arbitrary graph is a Hamiltonian graph or not. ∴ we need to analyse the graph to answer the question.



Topic : Dirac's theorem

- Let G = (V,E) be a graph with n vertices in which each vertex has degree at least n/2, Then G has a Hamiltonian cycle.

and hence 'G' is a Hamiltonian graph



Topic : Distance

- The distance between two vertices of graph G is defined as number of edges on the shortest Path {i.e. the path that contain minimum no. of Edges} between those two vertices.

distance b/w

a-b = 3

a-c = 4

a-d = 6

a-e = 1

a-f = 2

a-g = 3

a-h = 4

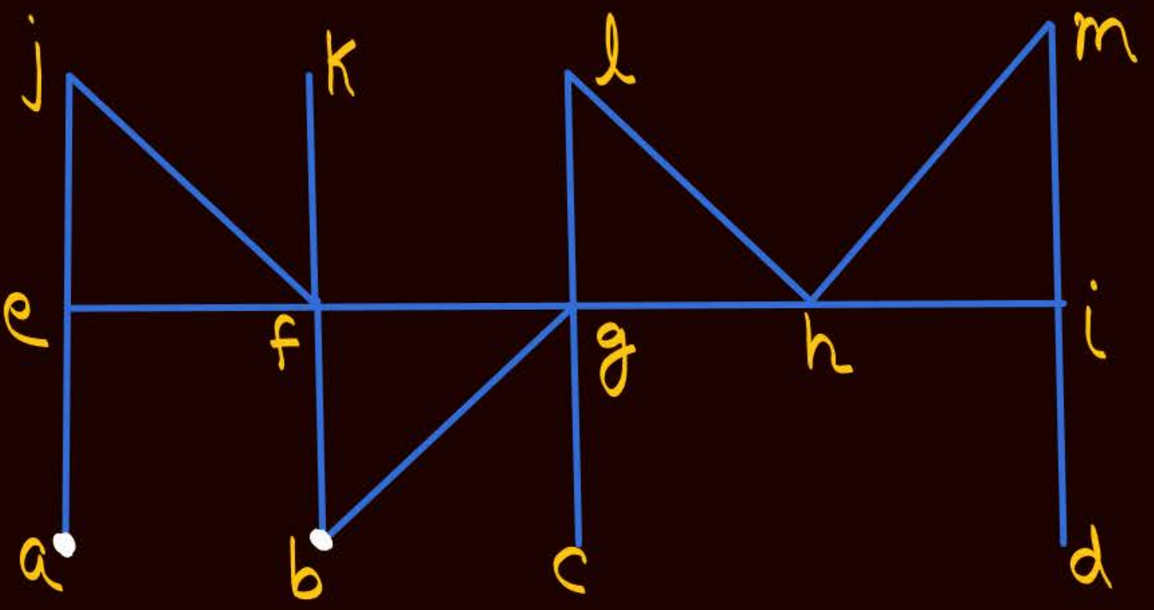
a-i = 5

a-j = 2

a-k = 3

a-l = 4

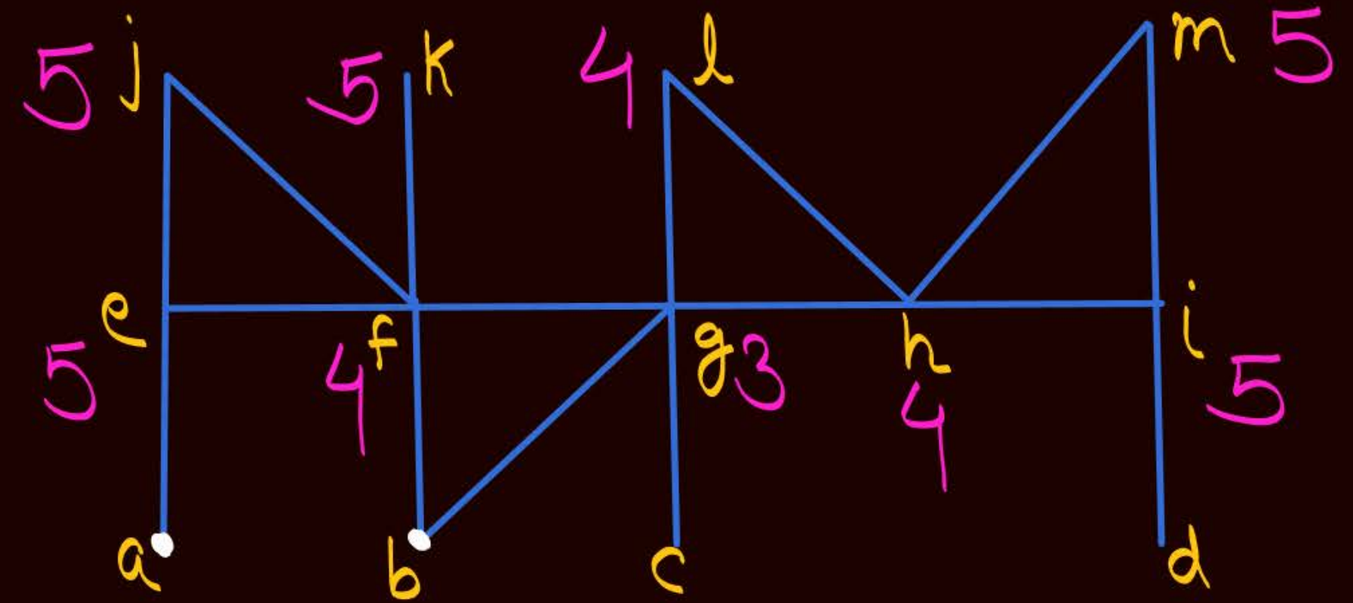
a-m = 5





Topic : Eccentricity

- Eccentricity of a vertex $V \in G$, is the maximum distance of vertex V from some other vertex of graph G.



Eccentricity = 6 4 4 6



Topic : Diameter

{ Diameter of graph G is the distance between two farthest vertices of graph G }

- Diameter of graph G is defined as maximum of the eccentricities of all the vertices of graph G.

  - In the above example,
    diameter of graph G = 6



Topic : Radius

Radius of graph G is defined as minimum of the eccentricities of all the vertices of graph. G.

- In the above example,

  Radius of graph G: 3



Topic : Centre

- The set of all vertices with minimum eccentricity is called Centre of the graph.
- In the above example,

  Centre of graph:- {g}



Topic : Centre

- The set of all vertices with minimum eccentricity is called Centre of the graph.
  - In the above example,
    Centre of graph:- {g}
- There may be more than one vertex with minimum eccentricity, in that case Centre will be defined using a set containing all vertices with minimum eccentricity



Topic : Girth

In graph theory, the girth of an undirected graph is the length of a shortest cycle contained in the graph. If the graph does not contain any cycles (that is, it is a forest), its girth is defined to be infinity.

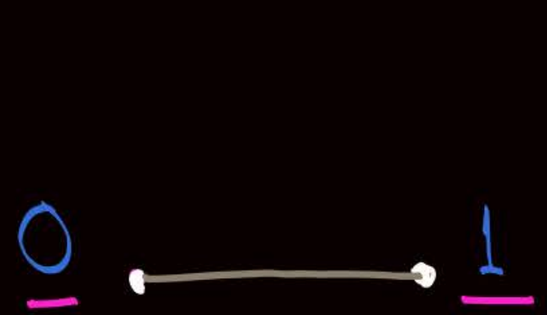


Topic : n-cube

- An n-cube graph, also known as a hypercube graph, is a graph where each vertex is represented by an n-bit binary string. Two vertices are connected by an edge if and only if their corresponding binary strings differ by exactly one bit.

There are '2^n' different n-bit binary strings {∴ 2^n vertices}

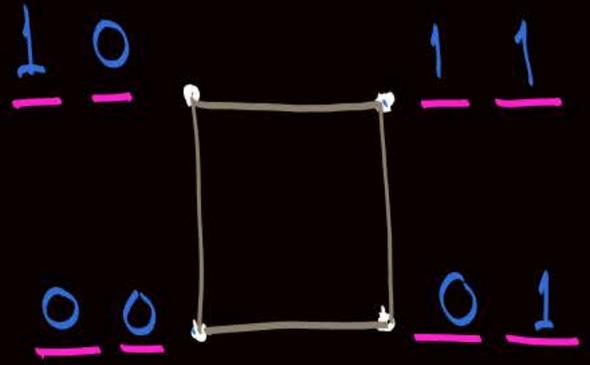
i.e. we can convert one into another by flipping exactly '1' bit



1-Cube

↓

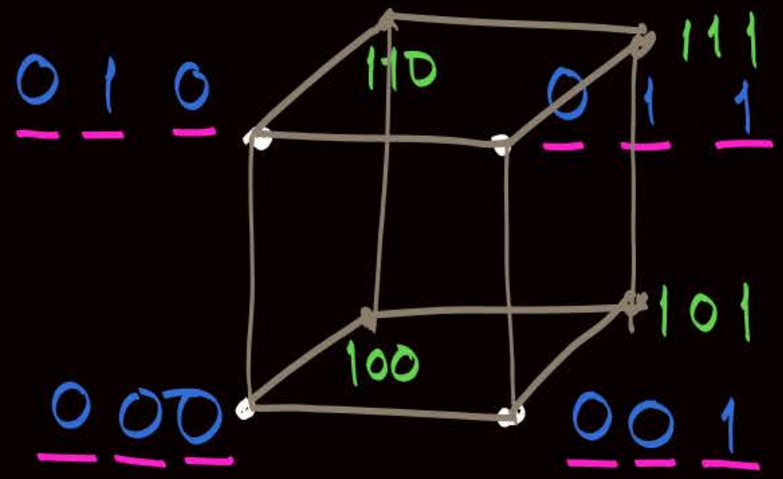
$2^1 = 2$ vertices



2-Cube

↓

$2^2 = 4$ vertices



3-Cube

↓

$2^3 = 8$ vertices

In a n-Cube graph,

No. of vertices $= 2^n$

& No. of Edges $= n \cdot 2^{n-1}$ ← same as boolean algebra

Note: ① Diameter of n-cube = "n"

② In an "n-cube" all cycles are of Even length.

∴ Every n-cube is a bipartite graph

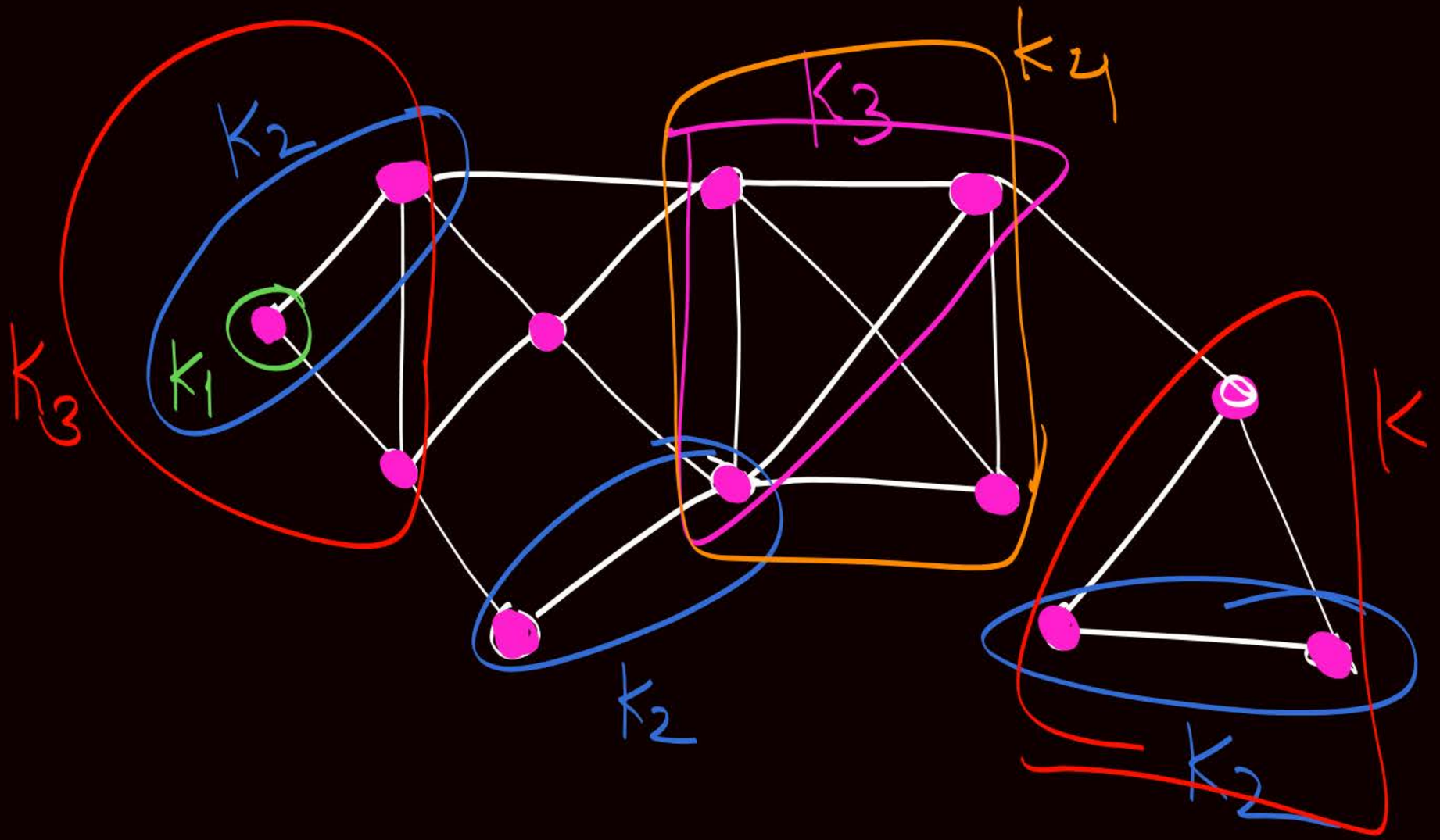
∴ $\chi(n\text{-cube}) = 2$

$n \geq 1$



Topic : Clique

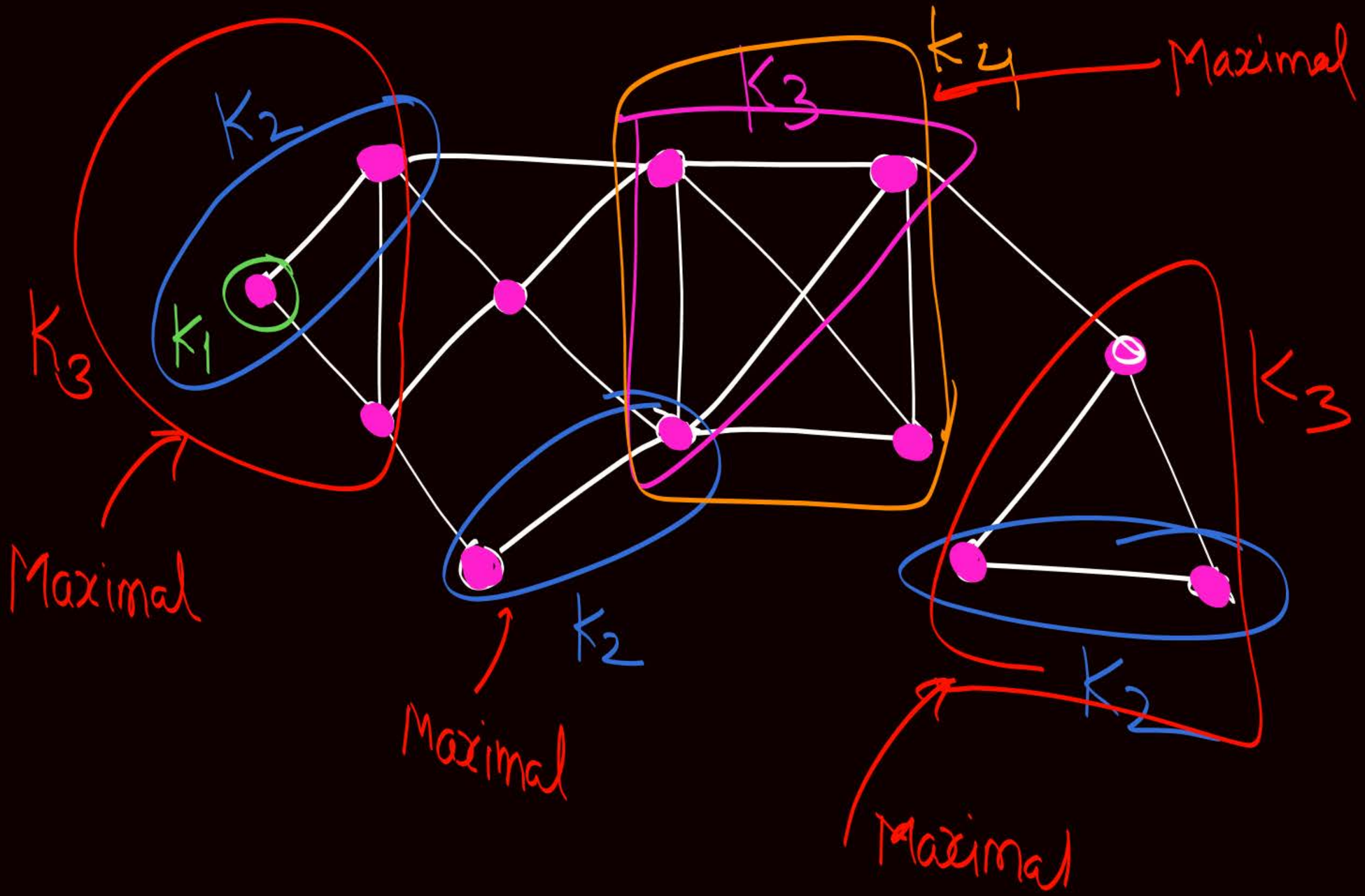
Any complete subgraph in graph G is called a clique.


K2
K3
K1
K3
K4
K2
K
K2



Topic : Maximal Clique

- A maximal clique is a clique that cannot be further expanded by including another vertex of the graph while maintaining the clique property.

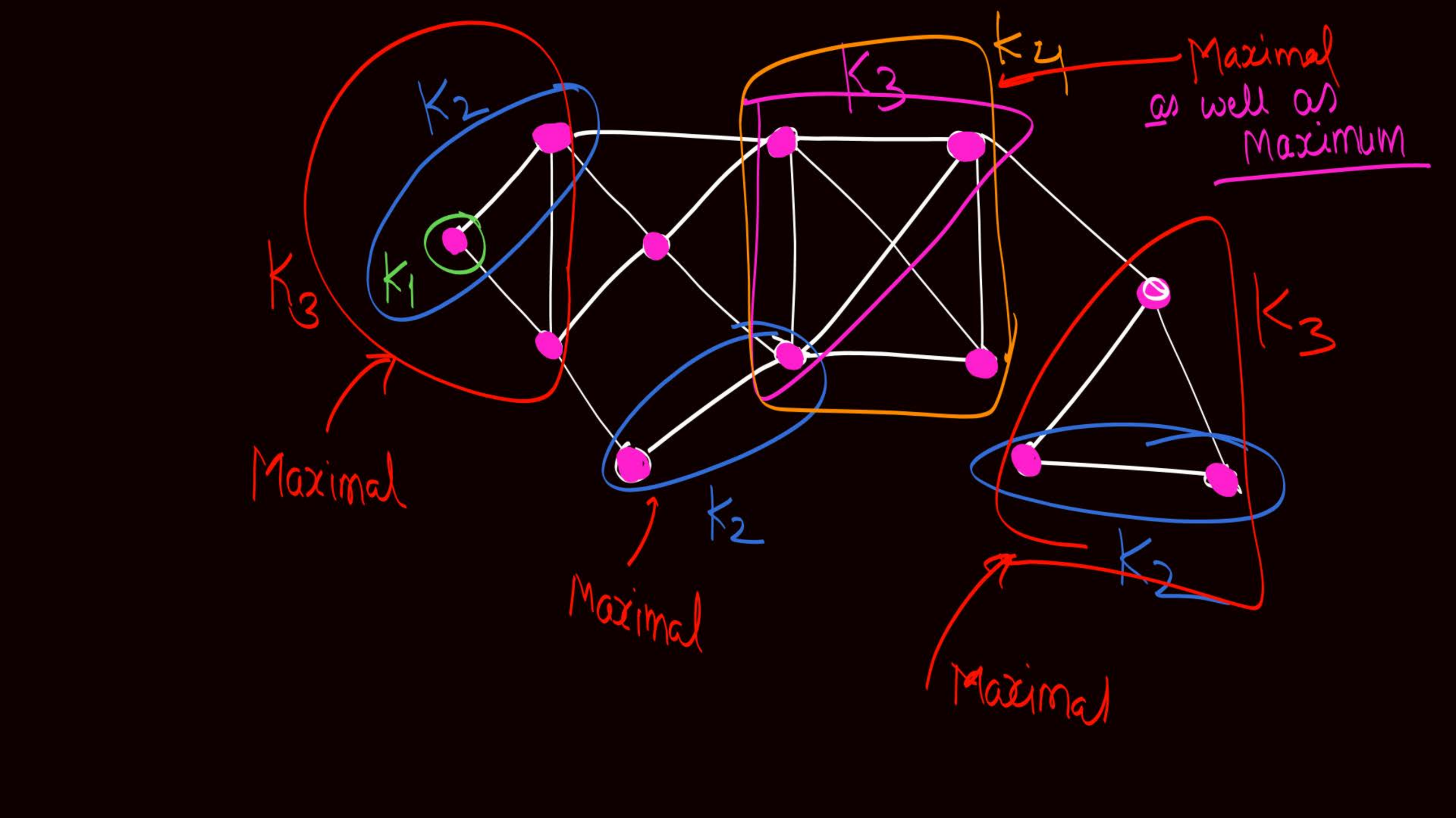

K2
K3
K1
Maximal
K3
K4
Maximal
K2
Maximal
K3
K2
Maximal



Topic : Maximum Clique

A maximum clique is a clique with the largest possible size within the graph

i.e. with maximum no. of vertices


K_2
K_1
K_3
Maximal
K_3
K_4
Maximal as well as Maximum
K_2
Maximal
K_3
K_2
Maximal



Topic : Clique number

Clique number of graph G is the number of vertices in maximum clique of graph G.

In above example,

Clique no. of graph G = 4



2 mins Summary

- Topic: Euler train, Euler circuit, and traversable graph
- Topic: Hamiltonian path and Hamiltonian cycle
- Topic: Distance, Eccentricity, Diameter, Radius and Girth
- Topic: n-cube graph
- Topic: Clique

Slide



THANK - YOU

Slide